

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Winter Examination-2023**

**Course: B. Tech.**

**Semester : VIII**

**Branch :** Biomedical Engineering / Electronics Engineering / Electronics and Telecommunication Engineering / Electronics and Communication Engineering

**Subject Code & Name: Introduction to Internet of Things(BTETPE801A)**

**Max Marks: 60**

**Date:15/01/2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                     | (Level/CO)    | Marks     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                              |               | <b>12</b> |
| <b>A)</b>   | What is SDN ? Explain the benefits of using SDN over IOT.                                                                                           | Apply         | <b>6</b>  |
| <b>B)</b>   | Write a note on Cloud Service Models.                                                                                                               | Analyze       | <b>6</b>  |
| <b>C)</b>   | Define Actuator and explain all the types of Actuator.                                                                                              | Remembering   | <b>6</b>  |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                              |               | <b>12</b> |
| <b>A)</b>   | Create Arduino code for Traffic Control System along with the circuit diagram.                                                                      | Apply         | <b>6</b>  |
| <b>B)</b>   | What is IoT ? and describe its characteristics and components.                                                                                      | Remembering   | <b>6</b>  |
| <b>C)</b>   | Explain COAP in detailed.                                                                                                                           | Understanding | <b>6</b>  |
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                              |               | <b>12</b> |
| <b>A)</b>   | What is activity monitoring in the IoT ? Explain its benefits and challenges.                                                                       | Understanding | <b>6</b>  |
| <b>B)</b>   | What is WSN ? and Explain its applications.                                                                                                         | Analyze       | <b>6</b>  |
| <b>C)</b>   | Write a short note on Bluetooth Technology.                                                                                                         | Apply         | <b>6</b>  |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                              |               | <b>12</b> |
| <b>A)</b>   | Write a short note on cloud computing.                                                                                                              | Understanding | <b>6</b>  |
| <b>B)</b>   | What is connective technology? and explain its impact on various industries.                                                                        | Apply         | <b>6</b>  |
| <b>C)</b>   | Explain the challenges faced during the deployment of Internet of Things.                                                                           | Analyze       | <b>6</b>  |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                              |               | <b>12</b> |
| <b>A)</b>   | Define Zigbee protocol . Explain Zigbee topologies, Application And Features of Zigbee Protocol.                                                    | Analyze       | <b>6</b>  |
| <b>B)</b>   | What is Unmanned Aerial Vehicle network? Explain its features and applications.                                                                     | Understanding | <b>6</b>  |
| <b>C)</b>   | Explain the following Operators in Python with example .<br>i) Bitwise Operators<br>ii) Comparison (Relational) Operators<br>iii) Logical Operators | Apply         | <b>6</b>  |

**\*\*\* End \*\*\***